

DD_DIS-OP-01 - Discount Runtime Application

Version: v1.0 **Bundle:** 05_billing **Status:** Draft for review **Primary owner:** Billing team **Related DDs:** DD_PLM-CFG-04, DD_SIP-03, DD_SIP-05, DD_BIL-01, DD_BIL-02-GEN-01, DD_BIL-05, DD_BIL-CFG-01, DD_EM-CFG-04, FOUNDATION_DROOLS

MVP simplification baseline: DIS-OP-01 starts with deterministic billing-time discount application for active assignments and simple campaign eligibility. Keep high-volume runtime logic efficient and testable; use Drools only for operator-configurable policy, not fixed arithmetic or invoice-line validation.

1. Purpose

DIS-OP-01 defines how SOPHIX applies configured discounts during billing, order charging, invoice generation, and wallet-credit operations.

The catalog and assignment modules answer these questions:

Question	Owning DD
What is this discount and how is it calculated?	PLM-CFG-04
Who has been granted this discount?	SIP-03
Which campaign may create discount assignments?	SIP-05
Which charge event is being billed?	BIL-CFG-01 and BIL-01
How is the discount applied at runtime?	DIS-OP-01

DIS-OP-01 is therefore the runtime calculation and audit layer. It receives a charge context from Billing, retrieves eligible active assignments from SIP-03, loads discount definitions from PLM-CFG-04, applies ordering/stacking/cap rules, returns discount lines to Billing, and records what happened.

Use this DD when:

- a recurring MRC charge must be reduced by a percentage or fixed amount;
- an installation charge must be waived during an acquisition campaign;
- a prorated invoice line must include discount impact;
- a prepaid wallet needs to receive a promotional credit;
- support must explain why a discount did or did not apply;
- a failed or disputed discount decision must be replayed without double-applying money.

Do not use DIS-OP-01 to create discounts or assign discounts to customers. That remains PLM-CFG-04 and SIP-03.

2. Scope And Boundaries

Capability	Owner
Discount definition, amount type, calculation type, validity, allowed event codes	PLM-CFG-04
Discount assignment to customer/account/subscription/order/franchise/campaign cohort	SIP-03
Campaign targeting and bulk assignment trigger	SIP-05
Runtime discount eligibility, calculation, stacking, capping, audit	DIS-OP-01
Charge creation and invoice item ownership	BIL-01 / BIL-02
Wallet debit/credit ledger	BIL-05
Billable event code catalog	BIL-CFG-01
Configurable approval for manual replay/override only	EM-CFG-04
Configurable runtime commercial rules	Drools

Boundary rules:

- DIS-OP-01 must not create or approve discount assignments.

- DIS-OP-01 must not mutate invoice totals directly. It returns discount lines to BIL-01/BIL-02 or posts a wallet credit through BIL-05 when discount type requires it.
- DIS-OP-01 must be idempotent for the same billable event and charge context.
- Runtime approval is not required for normal discount application. Approvals happen before assignment/campaign launch. EM-CFG-04 is used only for exceptional manual replay, override, or write-off style operations.

3. Runtime Position

```

flowchart LR
    A["BIL-01 receives billable event"] --> B["Build charge context"]
    B --> C["DIS-OP-01 preview/apply"]
    C --> D["SIP-03 active assignments"]
    C --> E["PLM-CFG-04 discount definition"]
    C --> F["Drools runtime policy"]
    C --> G["Discount decision and lines"]
    G --> H["BIL-01 charge result"]
    H --> I["BIL-02 invoice lines"]
    G --> J["BIL-05 wallet credit when needed"]

```

DIS-OP-01 should be callable in two modes:

Mode	Meaning	Used by
PREVIEW	Calculate expected discount without financial posting.	Quote, order capture, Backoffice, customer care.
APPLY	Calculate, persist, and return idempotent discount application lines.	BIL-01, BIL-02, BIL-05 integration.

4. Terminology

Term	Meaning
Charge context	Runtime payload describing the customer, subscription, package, event code, base amount, tax context, and date being charged.
Candidate assignment	Active SIP-03 discount assignment that may apply to the charge context.
Discount definition	PLM-CFG-04 record that defines amount, calculation type, validity, caps, event applicability, and stacking metadata.
Runtime decision	Auditable decision showing candidates, accepted discounts, rejected discounts, and reasons.
Discount application	The persisted result of applying discounts to one charge context.
Discount line	One monetary effect returned to Billing or Wallet.
Stacking	Whether multiple discounts can apply to the same charge context.
Cap	Maximum discount amount allowed per event, period, customer, assignment, or campaign.

5. Status Model

5.1 Discount Application Status

Status	Meaning	Next statuses
DRAFT	Runtime calculation started but not finalized.	APPLIED, REJECTED, FAILED
APPLIED	Discount lines were calculated and returned/posted successfully.	REVERSED
REJECTED	No discount was applied because eligibility or rule checks failed.	none
FAILED	Runtime error happened before final decision.	APPLIED, REJECTED by replay
REVERSED	Financial impact was reversed by an approved billing adjustment.	none

5.2 Runtime Decision Result

Result	Meaning
ACCEPTED	Candidate discount was used in final calculation.
REJECTED_NOT_ELIGIBLE	Candidate did not match package, event, channel, customer, geography, or date.
REJECTED_STACKING_CONFLICT	Candidate conflicted with a higher-priority discount.
REJECTED_CAP_REACHED	Assignment, campaign, or event cap was already reached.
REJECTED_ZERO_BASE_AMOUNT	Base amount was zero or negative and policy forbids discount.
REJECTED_RUNTIME_POLICY	Drools/runtime policy rejected the candidate.

6. Data Model

6.1 discount_application

What it is for: stores one runtime application attempt for a specific billable event and charge context.

When to use it: every APPLY request from BIL-01/BIL-02 that may produce discount lines. A PREVIEW request can optionally store a short-lived application record if audit is required, but normal preview should not persist unless configured.

Sample meaning: dapp-001 means the recurring MRC event for subscription SUB-700045 was charged on 2026-06-30, DIS-OP-01 evaluated it once, applied KES 300 in discount, and returned the result to Billing.

Column	Type	Purpose	Sample
application_id	UUID PK	Runtime discount application id.	dapp-001
operator_code	TEXT	Tenant/operator.	WIK
billable_event_code	TEXT	BIL-CFG-01 event code.	RECURRING_MRC
billable_event_ref	TEXT	Idempotent event ref from BIL-01/BIL-02.	bev-20260630-sub-700045
charge_context_hash	TEXT	Hash of the normalized charge context.	sha256:9f1b...
customer_id	TEXT NULL	Customer id.	CUS-100045
account_id	TEXT NULL	Account id.	ACC-900045
subscription_id	TEXT NULL	Subscription id.	SUB-700045
order_id	TEXT NULL	Order id for onboarding/order charges.	ORD-1001
package_ref	TEXT NULL	Package ref being charged.	pkg-fiber-100m
package_version_id	TEXT NULL	Package version charged.	pkv-100m-202607
currency_code	TEXT	ISO currency.	KES
base_amount	NUMERIC(14,2)	Amount before discount and tax.	3000.00
total_discount_amount	NUMERIC(14,2)	Sum of accepted discount lines.	300.00
net_amount	NUMERIC(14,2)	Base amount after discount, before any later tax handling.	2700.00
mode	TEXT	PREVIEW or APPLY.	APPLY
status	TEXT	DRAFT, APPLIED, REJECTED, FAILED, REVERSED.	APPLIED
decision_summary_code	TEXT	Main decision summary.	DISCOUNT_APPLIED
source_module_code	TEXT	Caller module.	BIL-01
idempotency_key	TEXT	Caller idempotency key.	discount-apply-bev-20260630-sub-700045
created_at	TIMESTAMP	Creation timestamp.	2026-06-30T00:05:00Z
finalized_at	TIMESTAMP NULL	Finalization timestamp.	2026-06-30T00:05:01Z

Recommended indexes:

- unique (operator_code, billable_event_ref, charge_context_hash, mode).
- (operator_code, subscription_id, created_at DESC).
- (operator_code, customer_id, created_at DESC).

- (operator_code, status, created_at DESC).

6.2 discount_application_line

What it is for: stores each accepted monetary discount effect inside one application.

When to use it: create one row per accepted discount assignment/definition pair. Billing consumes these rows to create invoice discount lines or reduced charge lines.

Sample meaning: da1-001 means assignment da-001 generated a KES 300 recurring service discount on one charge event.

Column	Type	Purpose	Sample
line_id	UUID PK	Discount line id.	da1-001
application_id	UUID	Parent application.	dapp-001
assignment_id	UUID	SIP-03 assignment id.	da-001
discount_code	TEXT	PLM-CFG-04 discount code.	RETENTION_10PCT_3M
discount_id	TEXT	PLM-CFG-04 discount id/version.	disc-1001
line_type	TEXT	PRICE_REDUCTION, FEE_WAIVER, WALLET_CREDIT, INVOICE_CREDIT.	PRICE_REDUCTION
calculation_type	TEXT	PERCENT, FIXED_AMOUNT, FULL_WAIVER, WALLET_CREDIT.	PERCENT
base_amount	NUMERIC(14,2)	Amount used for this calculation.	3000.00
discount_rate	NUMERIC(9,4) NULL	Percentage rate if applicable.	0.1000
discount_amount	NUMERIC(14,2)	Applied discount amount.	300.00
currency_code	TEXT	Currency.	KES
priority	INT	Runtime priority used.	100
stacking_group_code	TEXT NULL	Stacking group used.	RECURRING_MRC
cap_check_code	TEXT NULL	Cap rule that passed.	MONTHLY_ASSIGNMENT_CAP
wallet_transaction_id	TEXT NULL	BIL-05 transaction when line posts wallet credit.	wtxn-001
invoice_line_ref	TEXT NULL	BIL-02 invoice line after invoice generation.	invline-9001
metadata_json	JSONB	Extra calculation details.	{"rateSource":"catalog","rounded":true}

Recommended indexes:

- (application_id).
- (assignment_id).
- (discount_code).
- (wallet_transaction_id).

6.3 discount_runtime_decision

What it is for: stores accepted and rejected candidate decisions so care, billing, and audit teams can explain why a discount did or did not apply.

When to use it: for every candidate returned by SIP-03 effective-assignment lookup, and for policy-level rejection before candidate selection.

Sample meaning: drd-002 records that another active discount was found but rejected because it belonged to the same stacking group and had lower priority.

Column	Type	Purpose	Sample
decision_id	UUID PK	Decision id.	drd-001
application_id	UUID	Parent application.	dapp-001
assignment_id	UUID NULL	Candidate assignment.	da-001
discount_code	TEXT NULL	Candidate discount code.	RETENTION_10PCT_3M
decision_result	TEXT	Accepted/rejected result.	ACCEPTED

Column	Type	Purpose	Sample
reason_code	TEXT	Machine-readable reason.	ELIGIBLE_AND_HIGHEST_PRIORITY
message	TEXT	Human-readable explanation.	Applied 10 percent retention discount.
rule_package_version	TEXT NULL	Drools package/version used.	discount-runtime-v3.2
source_context_json	JSONB	Candidate facts and relevant context.	{"eventCode": "RECURRING_MRC", "packageRef": "pkg-fiber-100m"}
decided_at	TIMESTAMP	Decision timestamp.	2026-06-30T00:05:01Z

Recommended indexes:

- (application_id).
- (assignment_id, decision_result).
- (decision_result, reason_code).

6.4 discount_runtime_cap_usage

What it is for: tracks discount cap consumption where caps are enforced by assignment, campaign, customer, or event period.

When to use it: any discount definition or assignment includes a maximum amount/count by period. This table prevents repeated runtime calls from exceeding approved commercial limits.

Sample meaning: customer CUS-100045 has consumed KES 900 from a monthly cap of KES 1000 for assignment da-001.

Column	Type	Purpose	Sample
cap_usage_id	UUID PK	Cap usage id.	dcu-001
operator_code	TEXT	Tenant/operator.	WIK
cap_scope_type	TEXT	ASSIGNMENT, CAMPAIGN, CUSTOMER, SUBSCRIPTION, ORDER.	ASSIGNMENT
cap_scope_ref_id	TEXT	Scope id.	da-001
period_start	DATE	Cap period start.	2026-06-01
period_end	DATE	Cap period end.	2026-06-30
currency_code	TEXT	Currency.	KES
used_amount	NUMERIC(14,2)	Consumed discount amount.	900.00
used_count	INT	Consumed application count.	3
last_application_id	UUID	Last application that updated usage.	dapp-001
updated_at	TIMESTAMP	Last update timestamp.	2026-06-30T00:05:01Z

Recommended indexes:

- unique (operator_code, cap_scope_type, cap_scope_ref_id, period_start, period_end).
- (operator_code, period_start, period_end).

6.5 discount_runtime_replay

What it is for: records manual or system replay of failed discount runtime applications.

When to use it: only when a previous application failed or a billing support operator needs to recompute an audited result. Replays that can change money should require EM-CFG-04 approval when configured.

Sample meaning: drep-001 is a support replay that recalculated failed discount application dapp-001 after SIP-03 assignment cache lag was repaired.

Column	Type	Purpose	Sample
replay_id	UUID PK	Replay id.	drep-001
original_application_id	UUID	Application being replayed.	dapp-001
new_application_id	UUID NULL	New application result.	dapp-002
requested_by_user_id	TEXT	Requester.	usr-billing-002

Column	Type	Purpose	Sample
reason_code	TEXT	Replay reason.	CACHE_LAG_REPAIR
approval_request_id	TEXT NULL	EM-CFG-04 approval request if required.	apr-drep-001
approval_mode	TEXT NULL	EM-CFG-04 mode.	MULTI_STEP
status	TEXT	REQUESTED, PENDING_APPROVAL, APPROVED, REJECTED, COMPLETED, FAILED.	COMPLETED
created_at	TIMESTAMP	Creation timestamp.	2026-07-01T10:00:00Z
completed_at	TIMESTAMP NULL	Completion timestamp.	2026-07-01T10:03:00Z

Recommended indexes:

- (original_application_id).
- (approval_request_id).
- (status, created_at DESC).

7. APIs

All APIs require service authentication for module-to-module calls. User-facing preview/search calls are exposed through Backoffice/API Gateway with RBAC from EM-CFG-03.

7.1 Preview Runtime Discount

Used by order capture, sales app, Backoffice quote screens, and support.

```
POST /api/discount-runtime/preview
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "sourceModuleCode": "FUL-02",
  "mode": "PREVIEW",
  "billableEventCode": "INSTALLATION_FEE",
  "billableEventRef": "quote-ORD-1001-installation",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": null,
  "orderId": "ORD-1001",
  "packageRef": "pkg-fiber-100m",
  "packageVersionId": "pkv-100m-202607",
  "channelCode": "SALES_APP",
  "franchiseId": "fr-019",
  "chargeDate": "2026-07-01",
  "currencyCode": "KES",
  "baseAmount": 2500.00,
  "taxIncluded": false
}
```

Response:

```
{
  "applicationId": null,
  "status": "APPLIED",
  "baseAmount": 2500.00,
  "totalDiscountAmount": 2500.00,
  "netAmount": 0.00,
  "lines": [
    {
      "assignmentId": "da-free-install-001",
      "discountCode": "ACQ_FREE_INSTALL",
      "lineType": "FEE_WAIVER",
      "discountAmount": 2500.00,
      "reasonCode": "FULL_WAIVER"
    }
  ],
  "decisions": [
    {
      "assignmentId": "da-free-install-001",
      "decisionResult": "ACCEPTED",
      "reasonCode": "ELIGIBLE_FOR_ORDER_INSTALLATION"
    }
  ]
}
```

```
]
}
```

7.2 Apply Runtime Discount

Used by BIL-01/BIL-02. This endpoint persists the result and must be idempotent.

```
POST /api/discount-runtime/apply
Idempotency-Key: discount-apply-bev-20260630-sub-700045
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "sourceModuleCode": "BIL-01",
  "mode": "APPLY",
  "billableEventCode": "RECURRING_MRC",
  "billableEventRef": "bev-20260630-sub-700045",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": "SUB-700045",
  "orderId": null,
  "packageRef": "pkg-fiber-100m",
  "packageVersionId": "pkv-100m-202607",
  "channelCode": "SYSTEM",
  "franchiseId": "fr-019",
  "chargeDate": "2026-06-30",
  "currencyCode": "KES",
  "baseAmount": 3000.00,
  "taxIncluded": false
}
```

Response:

```
{
  "applicationId": "dapp-001",
  "status": "APPLIED",
  "baseAmount": 3000.00,
  "totalDiscountAmount": 300.00,
  "netAmount": 2700.00,
  "lines": [
    {
      "lineId": "dal-001",
      "assignmentId": "da-001",
      "discountCode": "RETENTION_10PCT_3M",
      "lineType": "PRICE_REDUCTION",
      "discountAmount": 300.00,
      "currencyCode": "KES"
    }
  ]
}
```

If the same idempotency key and same normalized charge context are received again, return the same result. If the same idempotency key is received with a different charge context hash, return 409 IDEMPOTENCY_KEY_CONFLICT.

7.3 Query Application History

```
GET /api/discount-runtime/applications?operatorCode=WIK&subscriptionId=SUB-700045&from=2026-06-01&to=2026-06-30
```

Used by Backoffice, billing support, reporting API, and customer care timeline.

7.4 Get Application Detail

```
GET /api/discount-runtime/applications/{application_id}
```

Returns application, lines, candidate decisions, cap usage updates, and source charge context snapshot.

7.5 Request Replay

```
POST /api/discount-runtime/applications/{application_id}/replay
Idempotency-Key: discount-replay-dapp-001-cache-lag
Content-Type: application/json
```

```
{
  "reasonCode": "CACHE_LAG_REPAIR",
  "requestedByUserId": "usr-billing-002",
  "expectedFinancialImpact": 300.00,
  "comment": "Assignment existed before billing but runtime cache was stale."
}
```

DIS-OP-01 calls EM-CFG-04 only when policy requires approval for replay. If approved, DIS-OP-01 recomputes and returns adjustment instructions to BIL-02/BIL-01 rather than editing old invoice rows directly.

7.6 Internal Approval Callback

Only used for replay/override approvals.

```
POST /internal/discount-runtime/replays/{replay_id}/approval-outcome
Content-Type: application/json

{
  "approvalRequestId": "apr-drep-001",
  "outcome": "APPROVED",
  "decisionComment": "Approved because customer should have received campaign discount."
}
```

8. Integration Contracts

8.1 Calls To SIP-03

DIS-OP-01 calls:

```
GET /api/discount-assignments/effective?operatorCode=WIK&subscriptionId=SUB-700045&eventCode=RECURRING_MRC&onDate=2026-06-
```

The response identifies assignments. DIS-OP-01 still validates the runtime charge context because the same assignment may be eligible for one event and not another.

8.2 Calls To PLM-CFG-04

DIS-OP-01 loads discount definitions by code/id. The definition must include:

- calculation type;
- value/rate;
- max amount and cap scope;
- allowed billable event codes;
- allowed package/service/category constraints;
- stacking group;
- rounding rule;
- wallet-credit behavior, if applicable.

8.3 Calls To BIL-05

For WALLET_CREDIT discount lines, DIS-OP-01 does not write wallet ledger rows itself. It calls BIL-05:

```
POST /api/wallets/{wallet_id}/credit
Idempotency-Key: wallet-credit-dapp-001-dal-001
Content-Type: application/json

{
  "amount": 500.00,
  "currency": "KES",
  "transactionType": "PROMOTIONAL_CREDIT",
  "reference": "dapp-001/dal-001",
  "reasonCode": "CAMPAIGN_WALLET_CREDIT"
}
```

8.4 Return To BIL-01/BIL-02

BIL-01 uses netAmount and lines to create charge results. BIL-02 uses returned lines to show invoice-level discount details. Tax calculation order must follow PLM-CFG-02 and BIL-02 tax rules.

9. Events

Event	Topic	When emitted	Consumers
DiscountApplicationComputed	sophix.billing.discount.application-computed	Preview or apply calculation completed.	REP, support timeline when configured.
DiscountApplied	sophix.billing.discount.applied	Apply produced at least one accepted discount line.	BIL-02, REP, CUST-INT.
DiscountApplicationRejected	sophix.billing.discount.rejected	No candidate applied.	REP, support diagnostics.
DiscountWalletCreditRequested	sophix.billing.discount.wallet-credit-requested	Wallet-credit discount was sent to BIL-05.	BIL-05, REP.
DiscountReplayRequested	sophix.billing.discount.replay-requested	Manual/system replay requested.	EM-CFG-04, audit.
DiscountReplayCompleted	sophix.billing.discount.replay-completed	Replay finished.	BIL, REP, support.

Use the transactional outbox pattern. Billing financial events must not be emitted from uncommitted database state.

10. Business Rules

Rule	Description	Reason
R-DIS-OP-01	Only active/effective SIP-03 assignments can be considered.	Prevent expired or unapproved discounts.
R-DIS-OP-02	Discount definition must be active and valid on charge date.	Prevent retired catalog use.
R-DIS-OP-03	Billable event code must be allowed by discount definition.	Avoid applying installation discount to MRC.
R-DIS-OP-04	Runtime target must match assignment scope.	Prevent customer/account/subscription leakage.
R-DIS-OP-05	Currency must match charge currency unless explicit conversion support exists.	Financial integrity.
R-DIS-OP-06	Stacking rules select accepted discounts by priority and group.	Revenue protection.
R-DIS-OP-07	Cap usage must be locked and updated atomically during apply.	Prevent cap overspend under concurrency.
R-DIS-OP-08	Discount amount must never make net amount negative unless the discount type is wallet credit or invoice credit and policy permits it.	Avoid invalid charge amounts.
R-DIS-OP-09	Repeated apply for same event/context must return the same result.	Idempotency.
R-DIS-OP-10	Same idempotency key with different context hash returns 409.	Protect against caller bugs.
R-DIS-OP-11	Preview must not update cap usage or wallet ledger.	Preview must be side-effect free.
R-DIS-OP-12	Wallet-credit discount must be posted through BIL-05 with its own idempotency key.	Ledger ownership.
R-DIS-OP-13	Manual replay that changes financial impact must call EM-CFG-04 if policy requires approval.	Governance.
R-DIS-OP-14	Old invoice rows are not edited by replay; adjustments are created through billing adjustment paths.	Audit and tax integrity.

11. Rule Engine Guidance

Use Drools for operator-variable decisions:

Rule file	Purpose
discount-runtime-eligibility.drl	Decide if candidate assignment is eligible for the runtime context.
discount-runtime-stacking.drl	Select which candidates survive stacking conflict checks.
discount-runtime-cap-policy.drl	Choose cap scope and cap period when policy differs by operator/channel/product.
discount-runtime-replay-approval.drl	Decide if a replay/override should call EM-CFG-04.

Do not use Drools for fixed validations such as JSON shape, required fields, date parsing, DB uniqueness, or idempotency hash comparison.

Example:

```
package com.sophix.discount.runtime

rule "Manual replay above threshold requires finance approval"
when
    $r : DiscountReplayFact(expectedFinancialImpact >= 1000.00, currencyCode == "KES")
    $d : ReplayDecision()
then
    $d.setApprovalRequired(true);
    $d.setApprovalPolicyCode("DISCOUNT_RUNTIME_REPLAY_FINANCE");
    $d.addReason("REPLAY_FINANCIAL_IMPACT_ABOVE_THRESHOLD");
end
```

12. Worker Model

DIS-OP-01 does not require a dedicated worker per action. For a budget-conscious medium deployment, consolidate these jobs inside the billing worker deployment:

Worker job	Purpose	Frequency
discount-cap-reconcile	Reconcile cap usage with application lines.	Nightly.
discount-application-outbox-publish	Publish outbox events.	Continuous/short interval.
discount-wallet-credit-retry	Retry failed BIL-05 wallet-credit calls.	Every 5 minutes.
discount-replay-runner	Execute approved replays.	On demand / queue.

Operational rule: one worker process can handle multiple job types if metrics and retry queues are separated.

13. Frontend And API Consumers

Frontend area	API/module used	Notes
Backoffice billing support	GET /api/discount-runtime/applications, GET /api/discount-runtime/applications/{id}	Explain applied/missed discounts.
Backoffice replay action	POST /api/discount-runtime/applications/{id}/replay plus EM-CFG-04 tasks	Only for authorized users.
Sales outdoor app	POST /api/discount-runtime/preview through FUL-02 quote flow	Used to show discount in quote.
Customer care CRM	history APIs plus CUST-INT timeline events	Explains invoice questions.
Reporting	events and read models	Discount cost by campaign, package, franchise, channel.

14. Infrastructure Recommendations

For a medium budget deployment, implement DIS-OP-01 inside the billing API/service deployment rather than a separate pod. Keep the database tables in the billing schema.

Component	Recommendation
API	Add /api/discount-runtime/* to billing API pods.
Database	PostgreSQL billing schema; use row locks for cap usage updates.
Cache	Redis cache for discount definition and active assignment read models, with short TTL and event invalidation.
Workers	Consolidated billing worker deployment.
Kafka	Use existing Kafka/MSK topics with transactional outbox.
Observability	Track discount latency, candidate count, rejection reason rate, cap lock contention, BIL-05 credit failures.

Suggested medium starting size:

Layer	Starting size
Billing API pods	2 replicas, 2 vCPU, 4 GB RAM each.
Billing workers	1-2 replicas, 2 vCPU, 4 GB RAM each.
PostgreSQL	Same HA billing DB cluster; no separate database.
Redis	Shared business cache cluster with namespace isolation.

15. AWS Deployment Notes

AWS layer	Recommendation
Compute	ECS or EKS service colocated with billing API and billing worker services.
Database	RDS PostgreSQL Multi-AZ if budget allows; otherwise single writer with strong backup policy in non-critical environments.
Cache	ElastiCache Redis for catalog/assignment cache and idempotency fast path.
Events	MSK or managed Kafka-compatible service; reuse existing billing topics.
Secrets	AWS Secrets Manager for service tokens and DB credentials.
Logs/metrics	CloudWatch plus Prometheus/Grafana if already deployed.

Use Graviton only after testing Java/KIE/runtime dependencies if Drools rules execute inside the same service. If Drools is called remotely through KIE Server, the billing service can run on ARM as long as its JVM/runtime dependencies are validated.

16. Testing Guidance

Test	Expected result
No active assignment	Application status REJECTED, no lines.
One active percentage discount	Correct amount and net amount returned.
Full fee waiver	Discount equals base amount, net zero.
Two discounts same stacking group	Highest priority accepted, other rejected with reason.
Two compatible stacking groups	Both accepted in configured order.
Cap reached	Candidate rejected with REJECTED_CAP_REACHED.
Concurrent applies near cap	Cap usage not exceeded.
Same idempotency key/context	Same response returned.
Same idempotency key/different context	409.
Wallet credit discount	BIL-05 credit called once with idempotency.
Preview request	No cap update, no wallet credit.
Replay requiring approval	EM-CFG-04 request created; no recompute until approved.

17. Acceptance Criteria

- Runtime preview and apply APIs exist.
- DIS-OP-01 reads effective assignments from SIP-03 and definitions from PLM-CFG-04.
- Billing apply is idempotent and records application, lines, decisions, and cap usage.
- Stacking, eligibility, cap, and replay approval rules are configurable through Drools where operator-variable.
- Wallet-credit discounts are posted through BIL-05, not direct DB writes.
- Replay/override financial changes use EM-CFG-04 when configured.
- Backoffice can explain why each candidate discount was accepted or rejected.
- Events are emitted through an outbox.
- Tests cover idempotency, cap concurrency, stacking, preview side effects, and wallet-credit integration.